Design of Enterprise IoT: Assignment 1b

Smart Turtle Breeding System (STBS)

Team 01

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Date: October 22, 2026

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Design scope. This report develops the A1a proposal into a component-level enterprise IoT design. It specifies commercially available field devices, the local and wide-area network, edge/fog and cloud processing, and end-to-end data flows. The device and connectivity profiles are reproduced from the supplied Excel templates so that the required A1b structure is retained.

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1 Task 1: Solution Overview

The Smart Turtle Breeding System (STBS) serves 20 indoor or semi-closed Chinese soft-shell turtle units. Each unit measures water temperature, pH, turbidity, ammonia, level and flow, while zone equipment monitors feed stock, humidity and video. Four local controllers preserve safe rules when the Internet is unavailable. A managed wired network carries controller telemetry and camera streams to an industrial edge computer, which validates data, detects events, buffers outages and performs video inference. Selected telemetry, alarms and evidence clips are sent securely to a cloud service for long-term records, dashboards and notifications. Operators review unit-specific conditions, feeding and nesting evidence, while technicians receive calibration and equipment-fault information. This design refines A1a by selecting deployable device models and separating deterministic local control, bandwidth-intensive edge analytics and cloud reporting (Munyaev et al., 2026; Prapti et al., 2022; Rastegari et al., 2023).

Word count: 133.

2 Task 2: IoT Devices

2.1 Overview and suitability

The device set favours wired, commercially available instruments because the units are fixed, wet and close to mains power. Industrial 4–20 mA and RS-485 probes tolerate longer runs and expose diagnostics; local PLCs provide bounded control and manual override. Temperature, pH, turbidity, ammonia, level and flow address high-value aquaculture risks, while a hopper sensor verifies feed availability. PoE cameras support activity and nesting evidence without battery maintenance. The selection links directly to the A1a value proposition: fewer routine rounds, earlier fault detection, more consistent feeding and auditable intervention, without making cloud availability a prerequisite for animal safety.

Word count: 91.

2.2 IoT Device Profile

The following pages are exported from the completed 'IoT Device Profile' worksheet in the supplied Excel workbook. The seven-column *Template-2_IoT-Device-Profile* structure is preserved. Each row now identifies a purchasable model, its measurable property, interface and supporting components, with the manufacturer URL shown in the characteristics field. The profile covers the field sensors and actuators used by the network and edge design in Tasks 3–5.

Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
The name of each IoT device	E.g., sensor, actuator, or device	What "thing" is being observed and/or acted upon?	Which aspect(s) of the "thing" is/are being observed and/or acted upon?	What are the characteristics of the device?	What are the components embedded in the device?	Why is the device selected for the proposed IoT solution?
DFRobot DS18B20 Waterproof Temperature Sensor (BOS0042) / 防水温度传感器	Sensor	Water in each breeding tank	Water temperature	Waterproof DS18B20 probe; 3–5 V; PH2.0-3P connector; sensing range −55 to 125 °C; display range −10 to 85 °C with ±0.5 °C stated error; requires a pull-up resistor. Source: DFRobot BOS0042.	Waterproof DS18B20 probe, 3-pin cable, pull-up resistor, PH2.0 adapter and controller input.	Direct temperature feedback supports heater control and alarm limits in wet tanks. The sealed probe is suitable for remote or humid measurement, while local validation prevents unsafe heating commands.
DFRobot SEN0189 Turbidity Sensor / 浑浊度传感器	Sensor	Water clarity in each breeding tank	Turbidity	Optical infrared emitter/receiver; 5 VDC; 40 mA maximum; response below 500 ms; selectable analogue 0–4.5 V or thresholded digital output; operating range −30 to 80 °C. The probe top is not waterproof.	Optical turbidity probe, interface board, A/D or GPIO input, analogue/digital selector and threshold potentiometer.	Provides an early warning of suspended solids, filtration faults and contamination. Digital threshold alarms can be local, while analogue values support trend analysis and water-change decisions.
DFRobot SEN0161-V2 Gravity Analog pH Meter / 模拟 pH 计 V2	Sensor	Water chemistry in each breeding tank	pH value	Signal board supply 3.3–5.5 V; analogue output 0–3.0 V; BNC electrode and PH2.0-3P signal connector; stated accuracy ±0.1 at 25 °C; electrode range pH 0–14, 5–60 °C, response below 2 min; 1 m cable.	Laboratory-grade pH electrode, V2 signal-conversion board, BNC cable, PH2.0-3P lead, calibration buffers and analogue input.	pH is a key water-quality variable for aquaculture. Two-point calibration and local plausibility/rate checks allow the PLC to alarm and bound corrective actions.
Lonhand LH-WQS-AN-485-UI NH3-N Ion Sensor / 氨氮离子传感器	Sensor	Water chemistry in each breeding tank	Ammonia-nitrogen concentration	Measurement range 0.1–18,000 mg/L; temperature range 0–60 °C; 4–20 mA and RS-485 outputs; IP68; 5 m cable; NPT3/4 or M39×1.5 connection; 0–3 bar; pH 4–10; repeatability ± 4%; PC/PBT corrosion-resistant housing.	Ion-selective ammonia electrode, 4–20 mA/RS-485 transmitter, 5 m cable, immersion fitting, PLC analogue/Modbus input and calibration solution.	Ammonia is a high-risk water-quality condition. A wired industrial output supports continuous trend logging, local alarm thresholds and dependable response even when the network is unavailable.
AXIS M3086-V Network Camera / 网络摄像机	Sensor	Breeding and nesting-zone activity and evidence	Video stream and event evidence	4 MP maximum resolution 2688×1512; 2.4 mm F2.1 lens; 130° horizontal and 93° vertical field of view; H.264/H.265/Motion JPEG; 25/30 fps; WDR up to 120 dB; Lightfinder; PoE Class 2; IP42 and IK08 variants.	Network camera, lens, PoE RJ45 connector, SD-card slot, control button, mounting dome and managed-switch port.	Video provides farm-wide context and event evidence for activity, feeding and nesting. PoE simplifies installation; edge analytics should retain only validated clips rather than upload continuous video.
Raspberry Pi 5 (8 GB)	Device	Edge gateway and local sensor aggregation	Local processing, protocol bridging and storage	Broadcom BCM2712 2.4 GHz quad-core 64-bit Arm Cortex-A76; 8 GB LPDDR4X-4267; standard 40-pin GPIO header; USB, Ethernet, Wi-Fi and Bluetooth; 5 V USB-C power and active cooling recommended for sustained workloads.	Raspberry Pi 5 board, microSD or SSD, 40-pin GPIO interface, USB/Ethernet/Wi-Fi links, power supply, case and active cooler.	A low-cost Linux edge node can bridge sensors, MQTT and dashboards, buffer data and pre-process camera events. Safety-critical interlocks remain in the dedicated PLC rather than the general-purpose board.
ESP32-WROOM-32	Device	Distributed sensor and low-power actuator nodes	Wireless telemetry and GPIO control	Dual-core Xtensa LX6 up to 240 MHz; 448 KB ROM, 520 KB SRAM and 4 MB SPI flash; 2.4 GHz 802.11 b/g/n up to 150 Mbps; Bluetooth 4.2/BLE; up to 32 GPIO; ADC, DAC, touch, PWM and serial peripherals; 3.0–3.6 V supply.	ESP32-WROOM-32 module, PCB antenna, 40 MHz crystal, flash, GPIO/ADC interface, regulated 3.3 V supply and enclosure.	Provides an economical wireless interface for non-critical distributed measurements. Wet-area deployments still need isolation, watchdogs and bounded local outputs before any actuator command is accepted.
Ubuntu Server 22.04	Device	Edge/server software platform	Container, storage, MQTT and API runtime	Ubuntu Server 22.04 LTS provides a maintainable Linux base for scale-out services, networking, storage, containers and system administration. The workload can run on Raspberry Pi 5 or another supported ARM/x86 host.	Ubuntu Server image, host storage, network interface, container runtime, MQTT broker, time-series database, API service and monitoring.	A documented LTS platform supports repeatable deployment, security updates and service separation for the edge gateway. It improves maintainability without placing deterministic safety control in the operating system.
HC-SR501 PIR Motion Sensor / 人体红外传感器	Sensor	Activity in breeding or nesting areas	Motion event	Pyroelectric dual-element PIR module; 4.5–20 V supply; below 50 µA stated current; 3.3 V high/0 V low GPIO output; cone below 100°; three pins; selectable repeatable/non-repeatable trigger and adjustable delay/sensitivity.	PIR element, Fresnel lens, comparator board, delay and sensitivity potentiometers, VCC/OUT/GND header and GPIO input.	A low-cost motion trigger can wake or tag camera recording and reduce continuous processing. It is not a safety sensor because initialization time, wind, lighting and animal movement can cause false or missed events.
PondMAX Solar Automatic Fish Feeder	Actuator	Feed hopper and pond feeding schedule	Scheduled feed dispensing	Solar charging with built-in 2200 mAh lithium battery; 1–6 feedings per day; adjustable S/M/L throw and 0–20 s feed duration; 3–10 mm food size; 12 V/1.66 A adapter specification in the supplied manual.	Solar panel, rechargeable battery, hopper, motor and spreader, timer/display controls, protective cover and mounting bracket.	Automated feeding improves schedule consistency and reduces manual labour. Feed commands should remain bounded by time, quantity and hopper checks, with an operator override and an audit record.
Bluefish MINI-200 Submersible Pump / 潜水泵	Actuator	Tank recirculation and water-change line	Water circulation and flow	12 VDC; 0.4 A; 4.8 W; 200 L/h open flow; 1.5 m maximum head; brushless motor; IP68 submersible construction; approximately 43×38×30 mm; below 35 dB; dry-run/block protection options.	Brushless pump, impeller, inlet/outlet, sealed motor, silicone vibration feet, 12 V supply, relay/MOSFET driver and flow feedback.	Low-voltage circulation supports filtration and controlled water changes. A relay, flow confirmation and dry-run protection prevent a blocked line or empty tank from damaging the pump.

3 Task 3: IoT Network Technology

The completed network-only worksheet is embedded below. The responses justify the field and plant technologies against range, power, data rate, latency, security, reliability and cost. The selected architecture uses industrial current loops, RS-485/Modbus RTU, pulse/UART, isolated 24 V control, managed Gigabit Ethernet/PoE and Ethernet-primary/LTE-failover WAN. The RS-485 and Ethernet limits are treated as engineering design constraints, not as a reason to put control-critical traffic on Wi-Fi (Modbus Organization, 2006; Teltonika Networks, n.d.; TP-Link, n.d.).

IoT devices, gateways, routers, etc. to connect	Main considerations	Specific questions	Responses
20 tank instrument sets; 4 × Arduino Opta RS485 controllers with A0602/D1608 I/O; 20 Reolink RLC-811A PoE cameras; TP-Link TL-SG3428XMP managed PoE+ switch; AAEON BOXER-8653AI edge computer; Teltonika RUTX11 router.	*Determine the connectivity between selected IoT devices, gateways, routers, etc.*	*Why do selected devices need to be connected with each other and to the network?*	Connectivity closes three loops: probes provide feedback to local control; controllers and cameras provide farm-wide context to the edge computer; selected events reach cloud users. It enables unit-specific alarms, verification of actuator outcomes, cross-unit comparison, nesting evidence and maintenance diagnostics. Safety interlocks remain in each Opta, so network loss reduces visibility rather than basic control.
	Determine communication and network technologies required to support connectivity	*What are the range and coverage requirements?*	Four RS-485 trunks each serve five nearby tanks and are deliberately kept below 100 m in a linear, terminated bus. Modbus guidance supports multidrop RS-485 and up to 32 unit loads per segment (Modbus Organization, 2006). Cat6 links from PLCs, cameras, edge computer and router remain within the 100 m copper-Ethernet channel limit. The control
		What are the power consumption constraints?	Mains is available, so continuity and protection matter more than battery life. A protected 24 VDC supply powers probes, PLCs and valves; PoE+ powers cameras; separate AC circuits feed heaters and pumps. A UPS supports PLCs, switch, edge computer and router for alarm delivery, buffering and orderly shutdown. Contactors, RCDs and manual isolation protect high-current loads.
		What is the required data rate (how much data needs to be transmitted over time)?	Minute telemetry and state changes remain below 0.1 Mbit/s. Cameras dominate: 20 H.265 streams at about 4–8 Mbit/s each produce 80–160 Mbit/s aggregate. A gigabit backbone provides headroom. The edge analyses lower-rate substreams, keeps main streams local and uploads only selected
		What are the latency requirements (how fast does data need to be transmitted)?	PLC interlocks and heater/valve decisions target sub-second local response. Local alarms and dashboard state should update within about 5 s. Cloud reports tolerate seconds to minutes. No safety action waits for a cloud round trip; cloud commands are authenticated requests that the edge and PLC revalidate against local limits.
		What are the security and reliability requirements?	The switch separates control, camera and management VLANs and applies ACLs, 802.1X/MAC controls and QoS. Its 24 PoE+ ports and 384 W budget cover 20 cameras with reserve (TP-Link, n.d.). Devices use unique credentials; unused services and camera P2P/UPnP are disabled; administration uses VPN. MQTT uses TLS and X.509 certificates. UPS power, watchdogs, local buffering, normally closed valves and Ethernet-primary/LTE-failover WAN improve resilience.
		What are the cost considerations?	*Wired installation and industrial probes cost more initially than hobby sensors or Wi-Fi, but reduce battery labour, interference and replacement risk in a wet fixed site. One managed PoE switch removes 20 camera power adapters. Four zone PLCs limit I/O cable length. Edge filtering avoids continuous video upload and*
		Which communication/network technologies are selected?	Field level: 4–20 mA for analogue probes; RS-485/Modbus RTU for multidrop sensors; pulse/UART for simple instruments; isolated relay/24 V control for actuators. Plant level: 1000BASE-T Cat6 and IEEE 802.3af/at PoE through a TP-Link TL-SG3428XMP (TP-Link, n.d.). WAN: Ethernet primary plus Teltonika RUTX11 dual-SIM LTE Cat 6 failover (Teltonika Networks, n.d.). Applications use local Modbus TCP/HTTPS/RTSP and outbound MQTT/TLS/HTTPS.

4 Task 4: Fog and Cloud Computing

The edge/fog and cloud continuation of the supplied connectivity template is embedded below. The Opta controllers execute local interlocks and bounded actuation; the BOXER edge node validates, correlates, buffers and analyses video; and AWS provides durable history, device identity, notifications and reporting. The cloud is therefore valuable for multi-user access and long-term analysis but is not placed in the immediate safety loop (AAEON, n.d.; Amazon Web Services, n.d.-a, n.d.-b, n.d.-c; Arduino, n.d.-b).

IoT devices and network devices (e.g., router, hub, gateway, etc.)	**Main considerations**	**Specific questions**	**Responses**
4 × Arduino Opta RS485 zone controllers; AAEON BOXER-8653AI industrial edge computer; TP-Link managed PoE+ switch; local operator web application; maintenance log/API.	*Determine the connectivity between IoT devices and network devices in the edge/fog layer*	*Is an edge/fog layer required? Why?*	Yes. Deterministic control must survive Internet loss, and 20 continuous video streams are too bandwidth-intensive and privacy-sensitive for routine cloud upload. Opta PLCs form the device edge; the BOXER is the farm fog node for protocol normalisation, quality checks, video inference, event correlation, short-term storage and store-and-forward delivery.
		Which network device(s) should handle computing across the edge/fog layer?	Each Opta executes five-tank control rules and exposes states over the isolated control VLAN. The AAEON BOXER-8653AI runs containerised acquisition, message brokering, rules, local dashboards and vision models. The managed switch and router forward and secure traffic but do not make animal-control decisions.
		What data is transmitted between the IoT devices and network device(s)?	Probes send temperature, pH, turbidity, ammonia, level, flow, feed distance and humidity with timestamps and diagnostics. PLCs add set-points, relay states, accumulated flow, manual/automatic mode, watchdog and alarm codes. Cameras send H.265 RTSP substreams and event windows. The edge adds device identity, tank/zone context and quality flags.
		What data processing and/or analysis tasks should be performed in the edge/fog layer?	PLCs scale signals, debounce pulses, check plausible ranges/rates, enforce hysteresis and time/volume bounds, and execute fail-safe control. The BOXER synchronises timestamps, detects stale or contradictory readings, aggregates summaries, correlates flow with level/valve state, infers activity/feeding/nesting candidates, queues messages during outages and
		If the fog connects with a user application, what is the purpose of the connection?	On the farm LAN, authorised users see current tank status, local alarms, event video, acknowledgement state and bounded manual controls even during WAN loss. The fog node labels unsynchronised data and preserves operational visibility and intervention capability.
		If the fog connects with a system, what is the purpose of the connection?	The fog service writes an append-only operations log and exports alarms, acknowledgements, maintenance flags, calibration due dates and actuator runtime through signed CSV/API. The maintenance system can create or close work orders but cannot issue raw heater, pump or valve commands.
AAEON BOXER-8653AI edge computer; Teltonika RUTX11 firewall/router; AWS IoT Core, time-series/object storage and rules; role-based web application; maintenance/work-order system.	*Determine the connectivity to the cloud*	*Is a cloud layer required? Why?*	Yes, but not for immediate safety. AWS provides durable off-site history, multi-user access, fleet identity, long-term comparison, reporting and notification integration. AWS IoT Core authenticates the edge through X.509 certificates, routes MQTT messages and maintains desired/reported state through device shadows (Amazon Web Services, n.d.).
		What data should be transmitted to the cloud, and is it from IoT devices or edge/fog layer devices?	The BOXER publishes validated minute summaries, state changes, alarms, acknowledgements, health/calibration metadata and daily aggregates by MQTT/TLS. Selected event thumbnails/clips and diagnostic bundles use HTTPS object upload. Raw continuous camera streams and high-frequency samples remain on site unless an authorised investigation requests them.
		What data processing and/or analysis tasks should be performed in the cloud?	IoT rules route telemetry to time-series storage and alarms to notification workflows; object storage retains approved evidence clips. Cloud jobs calculate cross-unit baselines, growth-stage comparisons, sensor-drift indicators and feed/energy/water summaries. Device shadows retain configuration state, while audit records capture user changes and
		If the cloud connects with a user application, what is the purpose of the connection?	A role-based web application provides historical charts, unit comparison, alarm notifications, evidence links, weekly reports and audit history. Users may request bounded set-point changes; the cloud records the request, but the edge/PLC rejects stale, unauthorised or out-of-range values.
		If the cloud connects with a system, what is the purpose of the connection?	The cloud exports approved maintenance tickets, calibration schedules, equipment runtime, water/feed consumption and compliance summaries through an API or scheduled CSV. Only necessary identifiers and evidence are included. Return status can close a ticket, but the external system cannot issue raw actuator commands.

5 Task 5: Overall Solution Connectivity

5.1 Graphical representation

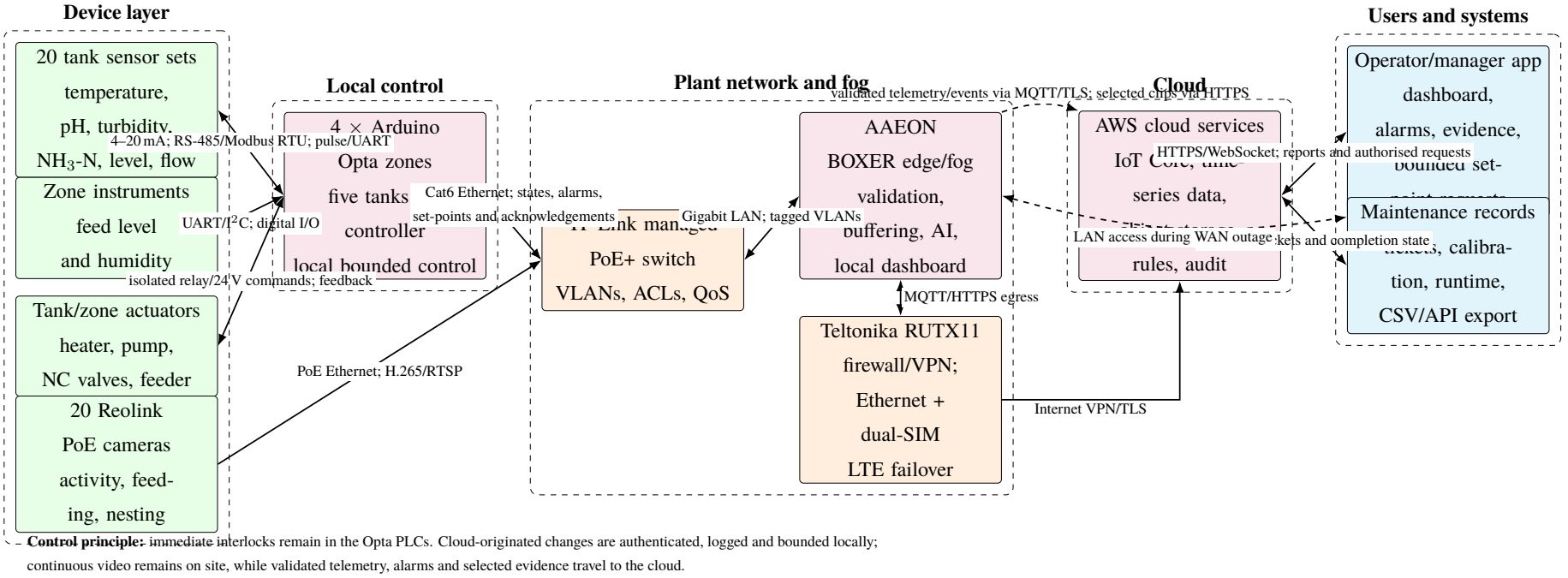


Figure 1: End-to-end STBS connectivity, processing locations and data flows.

5.2 Description of the diagram

Figure 1 separates safety control from analytics and reporting. Tank probes and actuator feedback terminate at four Opta PLCs, each responsible for five units. The PLCs execute bounded control locally and exchange states over the control VLAN. Twenty PoE cameras and the PLC uplinks reach a managed switch; the BOXER edge computer consumes separated streams, validates telemetry, runs video inference and buffers outages. The router exposes no field device directly and supplies secure primary-Ethernet/dual-SIM WAN service. The edge publishes validated summaries and events by MQTT/TLS and uploads only selected clips by HTTPS. AWS stores history, routes alarms and serves authenticated dashboards and maintenance exports. Users may acknowledge alarms or propose set-points, but every downward request is logged and revalidated at the edge and PLC. During Internet loss, local control, event storage and the LAN dashboard continue.

Word count: 128.

6 Task 6: Conclusion

The design fulfils the A1a plan by linking continuous water and feed measurements, bounded actuation, video evidence and role-specific alerts across 20 units. Industrial wired probes and zone PLCs improve repeatability and keep essential temperature, circulation and water-exchange rules independent of Internet availability. Edge processing contains video bandwidth, correlates commands with level/flow feedback and preserves a local operating view; cloud services add durable history, comparison, notification and maintenance reporting. The main limitations are probe fouling and calibration effort, the capital and installation cost of industrial sensors and protected wiring, and the need to train and validate vision models under changing light, water and turtle density. Ammonia and video inference are indicators rather than diagnoses, while feeder runtime only estimates delivered mass. A staged pilot on two units should validate thresholds, hydraulic timings, false-alarm rates, model performance and operating procedures before all 20 units are commissioned.

Word count: 139.

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